Aggregate Nulls, Compositional Shifts: AI Exposure and the Junior Employment Share in UK Occupations

Extended Essay

EC423 — Labour Economics

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Abstract

Todo.

1 Introduction

Since the launch of ChatGPT in autumn 2022, generative AI has undergone rapid development over the past three years. Beyond text generation, AI has achieved notable breakthroughs across a range of creative domains — including code generation, image synthesis, music composition, and video production. The pace of these advances has far exceeded the expectations of most observers, raising pressing questions about the implications for workers in occupations traditionally considered resistant to automation.

Early empirical evidence suggests that these advances are already reshaping labour markets for creative workers. Freelancers on Upwork in AI-exposed occupations experienced a 2 per cent decline in the number of contracts and a 5.2 per cent fall in earnings following the release of major generative AI tools, with image-related workers suffering considerably steeper losses (Hui et al., 2024).

A recent paper by Pabba (2026) argues that AI tools should not be treated merely as software, but rather as a form of capital.

Measuring the impact of AI is hard. Unlike discrete, large-scale shocks such as the COVID-19 pandemic, constructing a credible counterfactual for the impact of AI on the labour market is particularly challenging, precisely because AI adoption is gradual and diffuse rather than sudden (Massenkoff & McCrory, 2026).

Another focus about this essay is about unemployment. I want to use the QLFS data in UK to investigate the impact of AI on unemployment, and whether the impact is different for junior and senior workers within occupations.

2 Literature Review

3 Theoretical Framework

Autor & Thompson (2025) introduce an "expertise" framework which argue that the effect of the task automation depends not only on the quantity of tasks displaced, but also on the *expertise level* of those tasks relative to the remaining "task bundle". When the automation replace the *inexpert* task, the expertise requirement would rise, leading the wage goes up, and the employment would also falls due to fewer workers clear the higher threshold. When the automation replaces the *expert* tasks, the outcome reverses. This "expertise bifurcation" resolves the longstanding puzzle of why routine task-intensive occupations have experienced declining employment but not uniformly declining wages.

Central to their framework is Assumption 3 (occupational task bundling): *each worker in an occupation must perform* all *of its non-automated tasks, and conditional on meeting the expertise threshold, all workers are equally productive. This inseparability property is well suited to their between-occupation analysis, where the relevant variation lies across occupations with differing task compositions.* However, it rules out within-occupation heterogeneity — precisely the margin this essay wants to investigates.

In the following section. I retain the core insight of Autor & Thompson (2025) that occupations bundle tasks of varying expertise levels, but relax the inseparability requirement. What I do is that I introduce within-occupation task specialization: junior and senior workers allocate effort disproportionately to tasks matching their comparative advantage. This modification preserves the bundling structure while permitting the within-occupation compositional shifts documented in the subsequent empirical analysis. It is also reasonable because from the data I use, I have the seniority variable

which allows me to distinguish different workers in the same occupation or same company/firm.

3.1 Model Setup

Consider a single occupation in partial equilibrium. Output Y is produced by combining three task types via CES aggregation:

$$Y = \left[\beta_L y_L^{\frac{\sigma-1}{\sigma}} + \beta_H y_H^{\frac{\sigma-1}{\sigma}} + \beta_G y_G^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \tag{1}$$

where $\sigma > 0$ is the within-occupation task substitution elasticity and $\beta_L, \beta_H, \beta_G > 0$ are task share parameters summing to unity. For normalization, we let $\beta_L + \beta_H + \beta_G = 1$. The parameter σ governs how readily the occupation substitutes between task types; it is conceptually distinct from the between-occupation elasticity λ in Autor & Thompson (2025).

Following Autor & Thompson (2025), tasks fall into three categories:

- Task L: low-expertise activities, including routine operations that are readily automatable by AI.
- Task H: high-expertise activities, including context-dependent judgement, managerial decision-making, and other functions requiring accumulated experience.
- Task G: generic tasks that are common across occupations and performed by all worker types regardless of seniority.

The inclusion of a generic task follows the framework of Autor & Thompson (2025), and is motivated by a distinctive feature of the current AI wave: unlike the industrial automation of the twentieth century, which primarily displaced manual and blue-collar labour, generative AI disproportionately affects white-collar occupations — precisely those traditionally regarded as requiring substantial investment in education (Webb, 2020). The impact of AI on occupational task structures thus operates in a top-down manner, penetrating first into the cognitive and professional tasks that define high-skill roles.

Each task type is produced as follows:

$$y_L = A \cdot K + a_L \cdot n_J^L, \quad y_H = b_H \cdot n_S, \quad y_G = c \cdot n_J^G \tag{2}$$

In task L, AI capital K (rented at rate r) and junior labour n_J^L are perfect substitutes, with A denoting AI capability. Task H requires senior workers exclusively: the judgement-intensive nature of these tasks imposes an expertise threshold that junior workers cannot clear, consistent with hierarchical expertise in Autor & Thompson (2025). Task G is performed by **only** junior workers; while senior workers could in principle perform generic tasks, comparative advantage ensures they specialize entirely in task H in equilibrium.[1]

Junior workers allocate their labour between tasks L and G subject to $n_J^L + n_J^G = n_J$. At the occupation level, labour supply responds to wages with constant elasticity:

$$n_J = \varphi_J w_J^{\varepsilon}, \quad n_S = \varphi_S w_S^{\varepsilon} \tag{3}$$

where $\varepsilon > 0$ captures workers' ability to enter or exit the occupation in response to wage changes.

3.2 Equilibrium and Comparative Statics

The competitive firm's cost minimization yields three equilibrium conditions. Full derivations are provided in Appendix A.3.

No-arbitrage in task L. Since AI and junior labour are perfect substitutes, the firm employs both only when their unit costs are equalized:

$$w_J = \frac{a_L \cdot r}{A} \tag{4}$$

Junior wages are pinned entirely by AI's cost-effectiveness. As AI capability A rises, w_J falls — regardless of conditions in tasks H or G.

Internal allocation of junior labour. Equalizing the marginal revenue product of junior labour across tasks L and G yields a constant task output ratio $y_G/y_L = \gamma$, where $\gamma \equiv (\beta_G c/\beta_L a_L)^{\sigma}$ is a structural parameter independent of A. Although AI transforms

[1]This requires $w_S > w_J$ in equilibrium, which holds under the maintained assumptions. See Appendix A.2 for verification.

the production of task L internally, the ratio of generic to low-expertise output is invariant to AI capability because both tasks share the same marginal factor at wage w_J.

Senior augmentation via complementarity. Senior wages satisfy:

$$w_S = p \cdot \beta_H \left(\frac{Y}{y_H} \right)^{\frac{1}{\sigma}} \cdot b_H \tag{5}$$

The term $(Y/y_H)^{1/\sigma}$ embodies the key channel: as AI raises y_L and hence total output Y, while y_H remains unchanged, the marginal revenue product of task H — and therefore w_S — rises. This is the within-occupation analogue of the between-occupation augmentation result in Autor & Thompson (2025).

Comparative statics. An increase in AI capability A produces the following adjustment. First, w_J falls via Equation 4, reducing n_J through elastic labour supply as junior workers exit to other occupations. Within the occupation, junior workers partially reallocate from task L to task G, but total junior employment nonetheless declines. Second, the rise in y_L increases total output Y, raising the marginal product of task H via CES complementarity. Senior wages w_S rise and, through elastic supply, n_S increases. The combined effect — falling n_J and rising n_S — produces a decline in the junior employment share. The net effect on total occupation-level employment is theoretically ambiguous, depending on the magnitudes of σ and ε.

3.3 Testable Predictions

The model generates four predictions that I take to the data:

1. **Compositional shift:** In occupations with higher AI exposure, the junior employment share declines following the AI shock, as junior workers are displaced from task L and exit the occupation.
2. **Senior stability:** Senior workers in high-exposure occupations become more stable, with higher employment or lower transition rates, driven by CES complementarity raising their marginal product.

3. **Wage adjustment:** Junior wages decline and senior wages rise, though the magnitudes may be small if the occupation-level supply elasticity ε is large — a prediction consistent with adjustment occurring primarily through quantities rather than prices.
4. **Aggregate ambiguity:** The net occupation-level employment effect is indeterminate in sign, consistent with the aggregate null results documented in the existing literature.

The generic task channel provides an additional nuance: because junior workers retain a productive role in task G, displacement from high-AI occupations is *partial* rather than total. The junior employment share declines but does not vanish, consistent with the patterns documented in the empirical analysis that follows.

4 Data and Identification Strategy

The data I used here is the positions dataset provided by Revelio Labs, which contains detailed information on job postings in the UK. For analysis, I used data from 2019 to 2025, which allows me to capture the period before and after the release of ChatGPT tool in late 2022.

For each position, I observe the predicted salary, the O*NET occupation code, and other basic characteristics. Two additional variables are particularly relevant to this study. The first is an ordinal seniority variable, ranging from 1 to 7, that captures the hierarchical level of each position within an organisational structure. A detailed mapping is provided in Table 1.[2] I take this variable at face value throughout the analysis. Lichtinger & Hosseini Maasoum (2025) validate the measure by showing that the distribution of job-title keywords aligns closely with expected seniority distinctions and remains stable both before and after the diffusion of generative AI, confirming that it reliably captures positional hierarchy over time.

[2]A detailed description of the seniority classification methodology is available at: https://www.data-dictionary.reveliolabs.com/methodology.html#seniority.

Seniority	Level	Examples
1	Entry	Accounting Intern, Software Engineer Trainee, Paralegal
2	Junior	Account Receivable Bookkeeper, Junior Software QA Engineer, Legal Adviser
3	Associate	Senior Tax Accountant, Lead Electrical Engineer, Attorney
4	Manager	Account Manager, Superintendent Engineer, Lead Lawyer
5	Director	Chief of Accountants, VP Network Engineering, Head of Legal
6	Executive	Managing Director (Treasury), Director of Engineering (Backend Systems), Attorney (Partner)
7	Senior Executive	CFO, COO, CEO

Table 1: Revelio Labs seniority classification

The second is the position number, which records the chronological order of each position within a given individual's profile, with lower values corresponding to earlier roles. This variable enables the analysis of within-individual career mobility, which I exploit in a later section.

I then merge the AI exposure scores constructed by Eloundou et al. (2024) to the positions dataset via the O*NET occupation code. Based on the merged scores, I classify occupations into three categories: low exposure (below the 25th percentile), medium exposure (25th to 75th percentile), and high exposure (above the 75th percentile).[3]

4.1 Workers' Mobility

From the

4.2 研究 ChatGPT 的可靠性:

ChatGPT's release was unanticipated, precluding anticipatory adjustment by firms; the two-month window between release and the end of the baseline period is too short for measurable hiring effects to materialise.

[3]The exposure scores are publicly available at: https://github.com/openai/GPTs-are-GPTs.

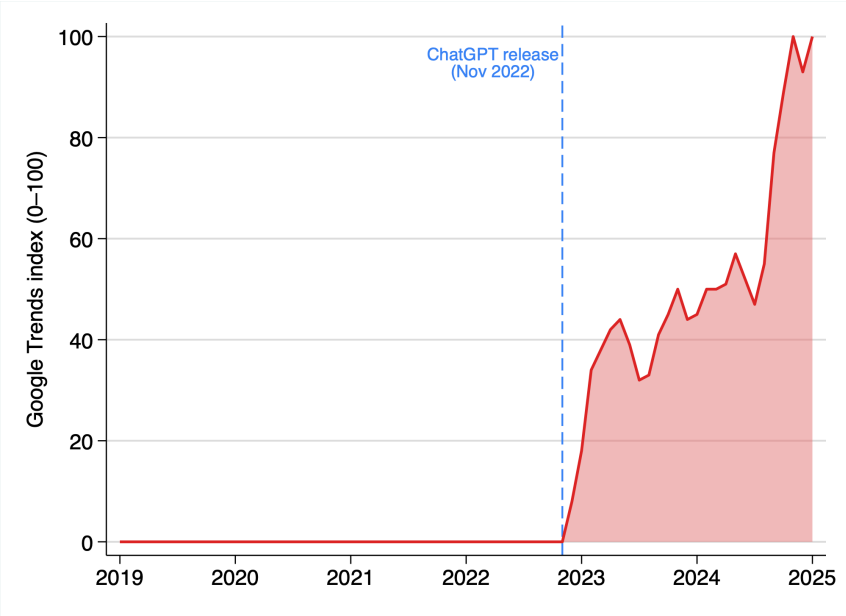

Figure 1: Google Trend Index on keyword "ChatGPT", 2019-2025

这个部分讨论一下 Identification Strategy 为草稿

Like the key research question from Heckman (1979), for those people who do not work, we couldn't observe wage. It is therefore necessary to address sample selection bias. In my essay, the intensive margin regression only look at those workers' outcome who successfully transit, but people who really transit isn't necessarily random. Imagine of two junior workers at high AI exposed jobs, one is with high ability one is not. During the post-ChatGPT period, those with better ability would transit to higher seniority jobs, and those weaker would stay or even drop out of the labour market. This would lead to higher Δ Seniority, it is not because AI helps junior, it is because only "winners" are observed.[4]

[4]The Heckman Model and the role of the inverse Mills ratio (Hazard ratio) in correcting for sample selection bias were covered in the EC423 Labour Economics lectures (Autumn Topic 1 and Autumn Topic 3), though the lecture didn't take a lot of part in Heckman two-step procedure. I apply the same logic here: the selection equation models entry into the transition sample, and the inverse Mills ratio corrects for non-random selection in the outcome equation. See Heckman (1979) for the original derivation.

5 Empirical Analysis

I begin by establishing the central compositional fact: the junior share of new hires in high-AI-exposure occupations declined significantly after ChatGPT, a finding consistent with that of Klein Teeselink (2025) at the firm level, he estimates a 5.8 per cent reduction in junior employment among highly exposed UK firms (King's College London, 2025). The contribution of this section is twofold. First, I provide a more rigorous identification check through event-study analysis with occupation-specific trend controls. Second, I lay the empirical foundation for the career mobility analysis that follows.

5.1 Compositional Shift

The present analysis complements this finding at a finer occupational granularity — individual position starts classified by O*NET code — and subjects it to a more demanding identification strategy through event-study estimation.

Figure 2 plots the monthly junior share of new position starts, separately for occupations above and below the median AI exposure score. Prior to late 2022, both series fluctuate around stable means: approximately 55 per cent for the high-exposure group and 85 per cent for the low-exposure group. The higher baseline share of junior workers in low-exposure occupations is unsurprising, as these roles tend to be more entry-level in nature. From early 2023 onward, the high-exposure series drops sharply, falling below 40 per cent by late 2025, while the low-exposure series exhibits only a modest decline. The divergence is visually striking and temporally aligned with the diffusion of generative AI.

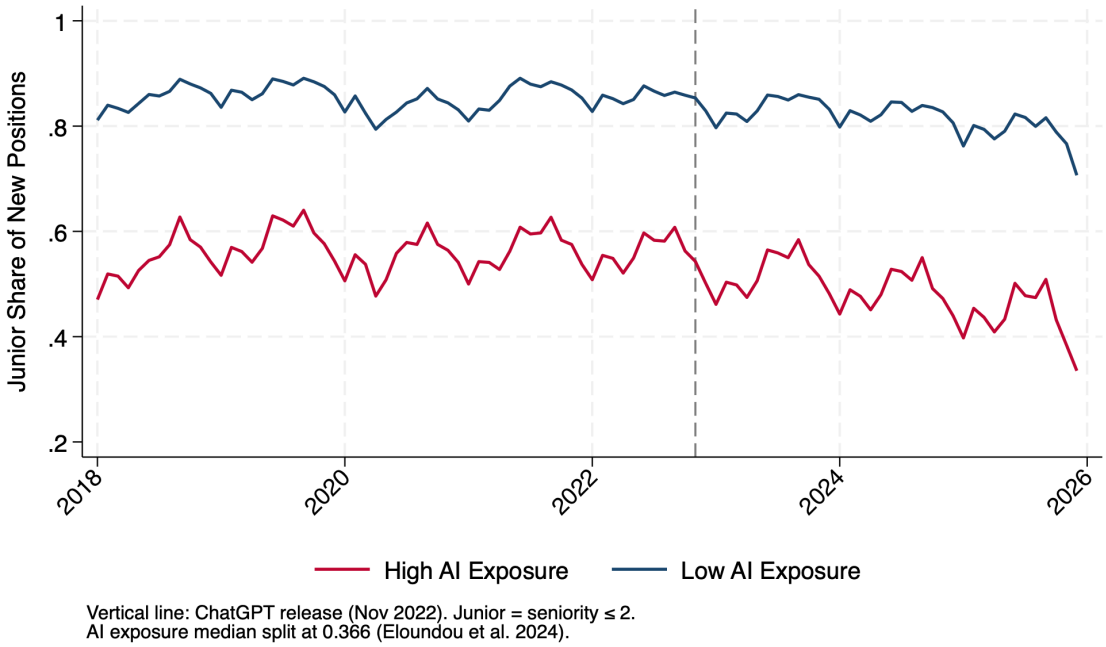

Figure 2: Monthly junior share of new position starts by AI exposure group. Junior defined as seniority ≤ 2 (Revelio Labs classification). AI exposure median split at 0.366 (Eloundou et al. (2024)). Vertical dashed line marks ChatGPT release (November 2022).

To quantify this divergence, I estimate a position-level difference-in-differences specification:

$$\text{Junior}_i = \alpha + \beta_1 \text{AIExp}_o + \beta_2 \text{Post}_t + \beta_3 \text{AIExp}_o \times \text{Post}_t + \varepsilon_i \tag{6}$$

where Junior_i is an indicator equal to one if position i has seniority level 2 or below, AIExp_o is the continuous AI exposure score of occupation o, and Post_t equals one for positions starting from January 2023, following Lichtinger & Hosseini Maasoum (2025), who document that the employment effects of generative AI emerge in the first quarter of 2023. Standard errors are clustered at the O*NET occupation level (917 clusters). The estimated interaction coefficient is $\hat{\beta}_3 = -0.087$ ($t = -4.78$, $p < 0.001$): after ChatGPT, occupations with higher AI exposure see a disproportionately larger decline in the share of new positions filled by junior workers. A hypothetical shift from zero to full exposure corresponds to an 8.7 percentage-point reduction; at the median exposure differential, this implies approximately 4.3 percentage points, consistent in magnitude with the firm-level estimates of Klein Teeselink (2025).

The pooled DiD estimate, however, assumes that pre-treatment trends in the junior share were parallel across exposure levels. I examine this assumption through a semi-annual event study:

$$\text{JuniorShare}_{o,t} = \alpha_o + \gamma_t + \sum_{k \neq 0} \beta_k \cdot \text{AIExp}_o \cdot \mathbb{1}[t = k] + \varepsilon_{o,t} \tag{7}$$

where α_o and γ_t denote occupation and half-year fixed effects, respectively, and the summation runs over all half-year periods with 2022H1 ($k = 0$) as the omitted baseline. This reference period is the last half-year that is both entirely pre-treatment and free from any early adjustment following the unanticipated release of ChatGPT on 30 November 2022. Observations are weighted by cell size.

Figure 3 presents the results. The pre-period coefficients ($k = -8$ to $k = -1$, spanning 2018H1 to 2021H2) fluctuate within a narrow band around zero, with no monotonic trend. Most estimates are individually insignificant: only three of the eight pre-period coefficients reach the 5 per cent significance level, and the largest absolute value is 0.017 (2020H2). A mild positive hump is visible during 2019H1–2020H2, plausibly reflecting differential COVID-era hiring in high-AI-exposure occupations, which are disproportionately white-collar and remote-compatible. These fluctuations are transient and subside before the treatment window: the coefficient at $k = -1$ (2021H2) is -0.007, statistically indistinguishable from zero. A formal test for a linear pre-trend, implemented by replacing the unrestricted pre-period interactions with a single occupation-specific trend term following Freyaldenhoven et al. (2019), yields an estimated slope of essentially zero ($\hat{\gamma} \approx 0$), confirming the absence of a systematic pre-existing divergence.

The post-period tells a sharply different story. The coefficient at 2022H2 — the half-year containing ChatGPT's launch — is -0.014, already statistically significant and suggestive of an initial response. From 2023H1 onward, the estimates decline steeply and near-monotonically: -0.059 at 2023H1, -0.096 at 2024H1, and -0.126 by 2025H2. All post-adoption coefficients are significant at the 1 per cent level, with 95

per cent confidence intervals well below zero. The post-period decline is an order of magnitude larger than the pre-period fluctuations and — unlike the pre-period hump — is monotonic and sustained over three years. A quarterly event study (see Appendix Figure 4) confirms that the decline begins precisely in 2022Q4–2023Q1, aligning with the onset documented by Lichtinger & Hosseini Maasoum (2025).

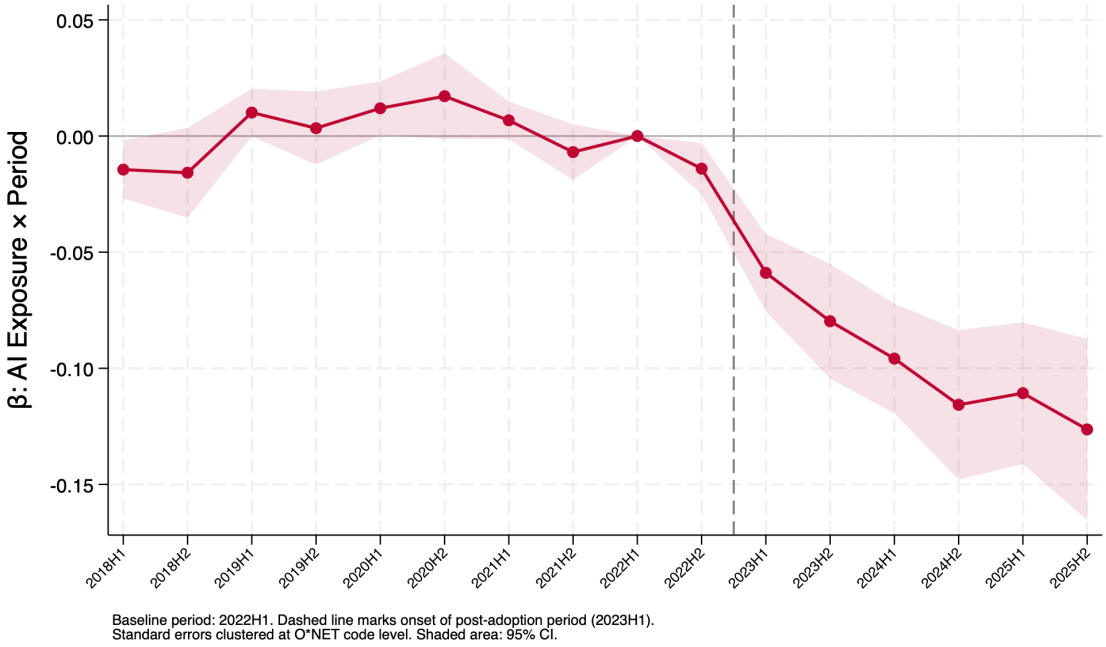

Figure 3: Event study: effect of AI exposure on junior share of new position starts. Each point represents the estimated coefficient $\hat{\beta}_k$ from Equation 7.

In summary, high-AI-exposure occupations experienced a sustained and statistically significant decline in the junior share of new hires following the diffusion of generative AI. By 2025H2, the estimated effect reaches approximately 13 percentage points for a fully exposed occupation. The magnitude of the post-period break, its monotonic trajectory, and its precise temporal alignment with the diffusion of generative AI collectively support a causal interpretation, notwithstanding the minor pre-period fluctuations. This compositional shift raises a natural follow-up question: what happens to the junior workers who would otherwise have entered these occupations?

5.2 Career Mobility

In this section, I turn to the analysis of career mobility, with particular emphasis on the post-ChatGPT period. The individual-level panel structure of the Revelio Labs data allows me to track career transitions directly. This addresses a limitation explicitly acknowledged by both Lichtinger & Hosseini Maasoum (2025) and Klein Teeselink (2025), who identify the analysis of individual career trajectories following AI exposure as a direction for future research.

For each transition, I construct outcome variables capturing the change in seniority level, the log wage change ($\Delta \ln w$), and an indicator for whether the worker switched O*NET occupation code. The core specification throughout is a triple-difference model:

$$\begin{aligned} Y_i = \beta_1 \text{Junior}_i + \beta_2 \text{AIExp}_i + \beta_3 \text{Post}_t + \beta_4 \text{Junior}_i \times \text{AIExp}_i + \\ \beta_5 \text{Junior}_i \times \text{Post}_t + \beta_6 \text{AIExp}_i \times \text{Post}_t + \beta_7 \text{Junior}_i \times \text{AIExp}_i \times \text{Post}_t + \varepsilon_i \end{aligned} \tag{8}$$

5.2.1 Extensive Margin: Who Moves?

Table 2 reports results from the extensive-margin analysis. Using the full sample of 27.5 million positions (including those without a subsequent transition), I estimate the triple-difference specification with an indicator for whether a transition is observed as the dependent variable. The estimate of β_6 is -0.373, indicating that senior workers in high-exposure occupations became substantially less mobile after ChatGPT, consistent with AI augmenting role-specific expertise and strengthening attachment to current positions. The triple-interaction term $\beta_7 = 0.228$ is positive and highly significant: relative to senior workers, junior workers in the same high-exposure occupations experienced a considerably smaller decline in transition probability. That is, while all workers in high-AI-exposure occupations became less mobile, this stabilisation was concentrated among senior workers; junior workers remained comparatively more exposed to turnover. The result is robust to excluding positions that began after July 2024 ($\beta_7 = 0.229$), ruling out mechanical right-censoring as a driver.

	Full sample	*Censoring-robust*
	(1)	(2)
Junior	0.194***	0.194***
	(0.038)	(0.038)
AI Exposure	0.365***	0.365***
	(0.053)	(0.053)
Post	0.222***	0.333***
	(0.031)	(0.029)
Junior × AI Exp.	–0.216***	–0.216***
	(0.053)	(0.053)
Junior × Post	–0.145***	–0.168***
	(0.030)	(0.028)
AI Exp. × Post	–0.373***	–0.352***
	(0.044)	(0.043)
Junior × AI Exp. × Post	**0.228*** **	**0.229*** **
	(0.042)	**(0.041)**
Constant	0.221***	0.221***
	(0.037)	(0.038)
N	27,472,206	24,531,424
R^2	0.009	0.022

Note: The dependent variable is an indicator for whether a position is followed by a transition to a different employer or occupation. Column (2) excludes positions with start dates after July 2024 to address potential right-censoring of ongoing positions. Cluster-robust standard errors in parentheses.

Table 2: Transition probability: triple-difference estimates

5.2.2 Intensive Margin: Transition Outcomes

Conditional on transitioning, Column (1) of Table 3 shows that junior workers leaving high-exposure occupations after ChatGPT gain significantly more seniority in their next position ($\beta_7 = 0.124$). However, this estimate is susceptible to positive selection: if AI disproportionately displaces weaker junior workers, those who successfully transition constitute a positively selected subsample, and the observed seniority gain overstates any genuine upskilling effect.

To address this concern, I implement a Heckman (1979) two-step correction. The selection equation models the probability of transitioning as a function of the main regressors and a firm-level exclusion restriction: the pre-period separation rate at the individual's company, which plausibly affects whether a worker transitions but not the characteristics of the destination position. The probit is estimated on a 10 per cent

random subsample for computational tractability, and the resulting inverse Mills ratio is included in the outcome equation estimated on the full sample.[5]

Column (2) reports the corrected estimates. The inverse Mills ratio is significant ($\hat{\lambda} = -0.174$), confirming that workers with lower predicted transition probability tend to experience smaller seniority gains conditional on transitioning — that is, positive selection into the transition sample is present. After correction, β_7 declines from 0.124 to 0.061, significant only at the 10 per cent level. This approximately 50 per cent reduction indicates that a substantial share of the observed seniority gain reflects compositional upgrading of the transition pool rather than a genuine treatment effect. Nevertheless, both interpretations reinforce the central thesis. If the effect is genuine, AI facilitates upward mobility for capable junior workers. If it reflects selection, AI filters out weaker junior workers from high-exposure occupations. Either channel implies that within-occupation composition shifts in the direction predicted by the model.

Columns (3) and (4) examine log wage changes upon transition. The triple-interaction term β_7 is essentially zero and statistically insignificant, indicating that junior workers are not differentially penalised in wage terms relative to senior workers. The adjustment mechanism thus operates primarily through the quantity margin — who remains in and who exits the occupation — rather than through differential wage compression.

Columns (5) and (6) examine occupational switching. The triple-interaction term is small and insignificant ($\beta_7 = -0.015$, $t = -1.00$ in the baseline). Junior workers do not systematically reallocate toward lower-AI-exposure occupations after ChatGPT. This null result is informative: the post-ChatGPT adjustment in career mobility operates along the seniority dimension rather than through occupational flight. Workers who transition move to positions of comparable AI exposure but at higher seniority — a

[5]The probit is estimated on observations where $\text{mod}(\text{user_id}, 10) = 0$. Coefficients are used to compute the linear index and the Mills ratio for all observations. Results are virtually identical when estimated on alternative subsamples ($\text{mod} = 1, 2, ...$).

pattern more consistent with upward filtering within related occupations than with wholesale retreat from AI-exposed work.

	Δ Seniority		*Δ ln Salary*		*Occ. Switch*	
	(1)	(2)	(3)	(4)	(5)	(6)
Junior	1.241***	1.197***	–0.042**	–0.042**	0.161***	0.180***
	(0.069)	(0.067)	(0.019)	(0.018)	(0.047)	(0.048)
AI Exposure	0.683***	0.591***	0.088***	0.087***	–0.143	–0.103
	(0.127)	(0.123)	(0.030)	(0.029)	(0.093)	(0.094)
Post	0.104***	0.063***	–0.032***	–0.033***	–0.004	0.013
	(0.017)	(0.018)	(0.009)	(0.008)	(0.009)	(0.009)
Junior × AI Exp.	–0.329***	–0.274**	–0.005	–0.004	–0.120*	–0.143**
	(0.109)	(0.107)	(0.030)	(0.029)	(0.072)	(0.073)
Junior × Post	–0.118***	–0.084***	0.039***	0.040***	–0.004	–0.019**
	(0.018)	(0.018)	(0.007)	(0.007)	(0.009)	(0.008)
AI Exp. × Post	0.052*	0.144***	–0.110***	–0.108***	0.019	–0.021
	(0.029)	(0.031)	(0.015)	(0.015)	(0.016)	(0.016)
Junior × AI Exp. × Post	**0.124*****	**0.061***	**–0.003**	**–0.004**	**–0.015**	**0.012**
	(0.034)	**(0.035)**	**(0.013)**	**(0.013)**	**(0.015)**	**(0.015)**
Δ Seniority			0.202***	0.202***		
			(0.003)	(0.003)		
IMR (λ)		–0.174***		–0.003		0.075***
		(0.019)		(0.005)		(0.011)
Constant	–0.801***	–0.612***	0.068***	0.071***	0.688***	0.606***
	(0.076)	(0.074)	(0.019)	(0.017)	(0.059)	(0.063)
N	12,487,869	12,487,869	10,676,912	10,676,912	12,487,869	12,487,869
R^2	0.138	0.139	0.341	0.341	0.020	0.022
Heckman correction	No	Yes	No	Yes	No	Yes

Table 3: Career mobility regressions: triple-difference estimates

6 Discussion

4.7: Discussion 部分仍然是想到哪写到哪，后期需要系统地进行整理和补充

Notice that research on the impact of AI on labour markets is still in its infancy, and there are many open questions about the mechanisms through which AI affects workers. One important argument is that we still couldn't find a consistent and convincing exposure measure for AI, which is crucial for empirical analysis. The most recent commonly used measurements can't broadly agree with each other on highly exposed

occupations, making it hard to draw robust conclusions about the impact of AI on labour markets (Gimbel et al., 2026).

这是一个有意识的简化。如果你想在 Discussion 里提一句，可以说“the model treats the junior-senior boundary as exogenous; in practice, AI tools may endogenously shift expertise thresholds within occupations, a channel explored at the between-occupation level by Autor and Thompson (2025) but left for future within-occupation analysis.” 这样既展示你意识到了这个 limitation，又合理地界定了你模型的 scope。

6.1 Why Wage Prediction “Fails”?

7 Conclusion

或许就像是 Restrepo (2025) 中描绘的美好愿景一样，我们将不再是推动世界前行的引擎，我们的劳作仅仅是高昂算力的暂时替代品。未来的社会必须直面的终极课题便是，当算力掌控了几乎所有的财富源泉之后，被挤出的不再只是 Junior，我们该如何通过类似于全民收入计划的行动来分配 AI 带来的红利，还是说我们将纵容财富不断集中，让它掌握在最“senior” 的一部分人手上呢？

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Appendix

A Derivation of Theoretical Model

A.1 Firm's Optimisation Problem and First-Order Conditions

The representative firm in the occupation takes output price p and factor prices w_J, w_S, r as given, and chooses AI capital M, junior labour in task L (n_J^L), junior labour in task G (n_J^G), and senior labour n_S to maximize profit:

$$\max_{M, n_J^L, n_J^G, n_S} \quad p \cdot Y - w_J\left(n_J^L + n_J^G\right) - w_S n_S - rM \tag{9}$$

where output Y is the CES aggregate defined in Equation 1, and the three task production functions are given by Equation 2.

Substituting the CES structure, the marginal revenue product of task $k \in \{L, H, G\}$ is:

$$p \cdot \frac{\partial Y}{\partial y_k} = p \cdot \beta_k \left(\frac{Y}{y_k}\right)^{\frac{1}{\sigma}} \tag{10}$$

The first-order conditions for an interior solution are:

$$p \cdot \beta_L \left(\frac{Y}{y_L}\right)^{\frac{1}{\sigma}} \cdot A = r \tag{11}$$

$$p \cdot \beta_L \left(\frac{Y}{y_L}\right)^{\frac{1}{\sigma}} \cdot a_L = w_J \tag{12}$$

$$p \cdot \beta_G \left(\frac{Y}{y_G}\right)^{\frac{1}{\sigma}} \cdot c = w_J \tag{13}$$

$$p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H = w_S \tag{14}$$

Derivation of the no-arbitrage condition. Dividing Equation 11 by Equation 12:

$$\frac{A}{a_L} = \frac{r}{w_J} \quad \Rightarrow \quad w_J = \frac{a_L \cdot r}{A} \tag{15}$$

Since AI and junior labour are perfect substitutes in task L, the firm employs both only when their unit costs are equalised. This condition pins junior wages entirely as a function of AI capability A and the rental rate r, independently of conditions in tasks H and G.

Derivation of the constant task output ratio. Dividing Equation 12 by Equation 13:

$$\frac{\beta_L (Y/y_L)^{\frac{1}{\sigma}} \cdot a_L}{\beta_G (Y/y_G)^{\frac{1}{\sigma}} \cdot c} = 1 \tag{16}$$

Simplifying:

$$\frac{\beta_L a_L}{\beta_G c} \cdot \left(\frac{y_G}{y_L}\right)^{\frac{1}{\sigma}} = 1 \quad \Rightarrow \quad \frac{y_G}{y_L} = \left(\frac{\beta_G c}{\beta_L a_L}\right)^{\sigma} \equiv \gamma \tag{17}$$

The ratio γ is a structural constant determined by technology and task share parameters. Intuitively, both tasks L and G are produced using the same marginal factor (junior labour at wage w_J), so their marginal costs move simultaneously. The CES demand system then fixes their output ratio regardless of A.

Senior wage expression. From Equation 14 directly:

$$w_S = p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H \tag{18}$$

This is the marginal revenue product of senior labour in task H.

A.2 Senior Specialization in Task H

The main text asserts that senior workers specialize entirely in task H in equilibrium despite being physically capable of performing task G. I verify this claim here. The intuition is simple.

If a senior worker were reallocated to task G, her marginal revenue product would equal that of junior workers in task G, which by Equation 13 is exactly w_J. Her marginal

revenue product in task H is w_S. Senior specialization in task H is therefore optimal whenever $w_S > w_J$, that is:

$$p \cdot \beta_H \left(\frac{Y}{y_H}\right)^{\frac{1}{\sigma}} \cdot b_H > \frac{a_L \cdot r}{A} \tag{19}$$

This condition holds when task H is sufficiently valued relative to task L — specifically, when the expertise-intensive component of the occupation commands a premium over routine tasks. This is empirically well supported: in the Revelio Labs data, salaries for senior positions (seniority ≥ 5) substantially exceed those for entry-level positions (seniority ≤ 2) within the same O*NET occupation. I maintain $w_S > w_J$ as a regularity condition throughout.

A.3 Comparative Statics

I now try to derive the effects of an increase in AI capability A on the equilibrium allocation, which covers mainly four steps.

Step 1: Junior wage and employment

From the no-arbitrage condition:

$$\frac{\partial w_J}{\partial A} = -\frac{a_L r}{A^2} < 0 \tag{20}$$

Substituting into the supply function (Equation 3):

$$n_J = \varphi_J \left(\frac{a_L r}{A}\right)^{\varepsilon} \quad \Rightarrow \quad \frac{\partial n_J}{\partial A} = -\varepsilon \varphi_J (a_L r)^{\varepsilon} A^{-(\varepsilon+1)} < 0 \tag{21}$$

An increase in AI capability monotonically reduces both the wage and employment of junior workers in the occupation. The magnitude of the employment response is governed by the supply elasticity ε: when ε is large, even a small decline in w_J induces substantial exit. Pinning down the precise value of ε is beyond the scope of this essay; the key qualitative prediction — that junior employment falls with AI capability — holds for any $\varepsilon > 0$.

Step 2: Task output levels

To characterize how task outputs respond to A, I reduce the model's dimensionality. Since $y_G = \gamma y_L$ with γ constant, I define $\tilde{\beta}_L \equiv \beta_L + \beta_G \gamma^{\frac{\sigma-1}{\sigma}}$ and write the CES aggregate as a function of two effective task outputs:

$$Y = \left[\tilde{\beta}_L y_L^{\frac{\sigma-1}{\sigma}} + \beta_H y_H^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}} \tag{22}$$

From Equation 11, the shadow price of task L output satisfies:

$$p \cdot \beta_L \left(\frac{Y}{y_L}\right)^{\frac{1}{\sigma}} = \frac{r}{A} \tag{23}$$

This implicitly defines the demand for y_L as a function of y_H and A. Holding y_H fixed, an increase in A reduces the effective marginal cost of task L output (r/A), which by CES demand raises y_L. Since $y_G = \gamma y_L$, task G output rises proportionally.

To verify that total output Y rises: from Equation 22, Y is increasing in both y_L and y_H. Since y_L increases and y_H is (initially) unchanged, Y unambiguously increases.

Step 3: Senior augmentation

I establish that the demand curve for senior labour shifts outward when A increases, and that the equilibrium senior wage and employment both rise.

Define the inverse demand for senior labour, holding A fixed:

$$w_S^D(n_S; A) \equiv p \cdot \beta_H \left(\frac{Y(n_S, A)}{b_H n_S}\right)^{\frac{1}{\sigma}} \cdot b_H \tag{24}$$

where $Y(n_S, A)$ is total output evaluated at the optimal task allocation for given n_S and A.

Claim 1: w_S^D is decreasing in n_S. An increase in n_S raises $y_H = b_H n_S$ and hence Y. However, y_H rises proportionally with n_S, while Y exhibits diminishing returns to y_H under the CES structure ($\sigma < \infty$). Consequently, $Y/(b_H n_S)$ falls, and so does w_S^D.

Claim 2: w_S^D is increasing in A. Holding n_S fixed, an increase in A reduces the marginal cost of task L, raising y_L and $y_G = \gamma y_L$. Since $y_H = b_H n_S$ is unchanged, Y increases while y_H does not, so $Y/(b_H n_S)$ rises and w_S^D rises.

Equilibrium effect. The demand curve $w_S^D(n_S; A)$ is downward-sloping and shifts outward when A rises. The supply curve $n_S = \varphi_S w_S^{\varepsilon}$ is upward-sloping. Standard com-

parative statics of supply-demand intersection imply that the new equilibrium features both higher w_S and higher n_S:

$$\frac{\partial w_S}{\partial A} > 0, \quad \frac{\partial n_S}{\partial A} > 0 \tag{25}$$

This confirms the augmentation result: AI capability improvements raise senior wages and senior employment via CES task complementarity.

Step 4: Compositional shift

Define the junior employment share as $s_J \equiv n_J/(n_J + n_S)$. From Steps 1 and 3, $\partial n_J/\partial A < 0$ and $\partial n_S/\partial A > 0$. Both the numerator and denominator of s_J are affected, the junior employment share unambiguously declines.

Internal reallocation of junior labour

Although total junior employment falls, the allocation of remaining juniors between tasks L and G also shifts. From $y_G = \gamma y_L$ and $y_G = c \cdot n_J^G$:

$$n_J^G = \frac{\gamma y_L}{c} \tag{26}$$

Since y_L is increasing in A (Step 2), n_J^G increases. Combined with declining total junior employment (n_J falls), this implies:

$$n_J^L = n_J - n_J^G \tag{27}$$

which falls faster than n_J itself. Junior workers are squeezed out of task L on two margins: AI directly substitutes their labour, and the remaining juniors reallocate toward task G where AI cannot substitute.

The above derivations assume an interior solution in which $n_J^L > 0$ For sufficiently large A, all junior workers exit task L and produce only task G output. I focus on the interior case, which is empirically relevant given that junior workers remain present in high-exposure occupations throughout the sample period.

A.4 Net Occupation-Level Employment

The net effect of AI capability on total occupation-level employment $N = n_J + n_S$ is:

$$\frac{\partial N}{\partial A} = \frac{\partial n_J}{\partial A} + \frac{\partial n_S}{\partial A} \tag{28}$$

The first term is unambiguously negative and the second is unambiguously positive (from my derivation in Appendix A.3). The net sign depends on the relative magnitudes, which are governed by the structural parameters. Senior augmentation dominates ($\partial N/\partial A > 0$) when:

- The task share of high-expertise tasks (β_H) is large, amplifying the complementarity channel.
- Senior supply is highly responsive (φ_S large or ε large), so that wage increases translate into large employment gains.

Conversely, junior displacement dominates ($\partial N/\partial A < 0$) when β_L is large, junior supply is highly responsive (φ_J large or ε large), and the initial junior share is high.

B Validation from QLFS Data

I also use data set from Quarterly Labour Force Survey by UK government. And I am also aware that the dataset itself exists.

这一步不确定会不会做，待定。

C More Evidence from Empirical Analysis

C.1 Event study at quarterly frequency

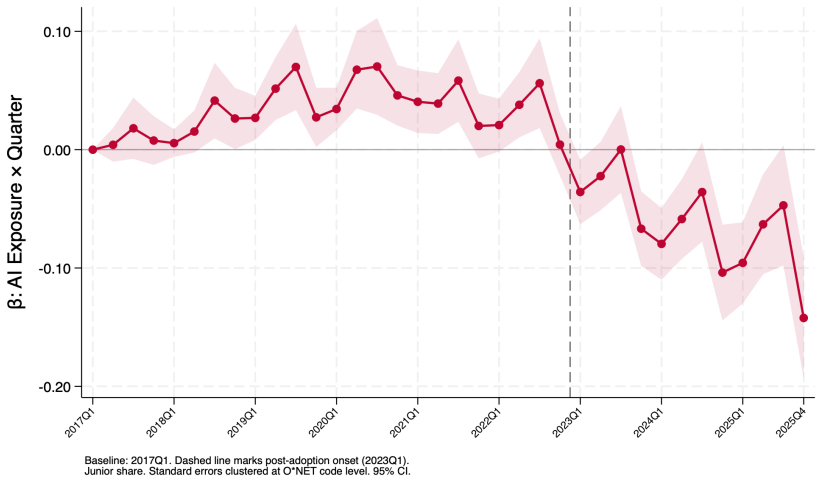

Figure 4: Event study: effect of AI exposure on junior share of new position starts on quarterly basis